

TTS Implementation

Setup:

```
dependencies:  
  flutter:  
    sdk: flutter  
  
  # for Supabase connection  
  supabase_flutter: ^2.8.0  
  
  # For Pose Detection  
  google_mlkit_pose_detection: ^0.10.0  
  google_mlkit_commons: ^0.6.1  
  
  # Hardware  
  camera: ^0.11.0  
  
  # For Text-to-speak  
  flutter_tts: ^4.0.2  
  path_provider: ^2.1.2  
  
  # For Videos  
  video_player: ^2.8.1  
  
dev_dependencies:  
  flutter_test:  
    sdk: flutter  
  flutter_native_splash: ^2.4.0
```

Figure 1: Integrating the Flutter TTS library to bridge Dart code with the Android Native Speech Engine

```

// For Text-To-Speech
import 'package:flutter_tts/flutter_tts.dart';

class TTSService {
  final FlutterTts _flutterTts = FlutterTts();
  // Guard to prevent overlapping voices
  bool _isSpeaking = false;

  TTSService() {
    _initTts();
  }

  void _initTts() {
    _flutterTts.setLanguage("en-US");
    _flutterTts.setPitch(1.0);
    // speaking speed
    _flutterTts.setSpeechRate(0.5);

    // Listen for when the app finishes speaking
    _flutterTts.setCompletionHandler(() {
      _isSpeaking = false;
    });
  }

  Future<void> speak(String text) async {
    // Only speak if it's not currently busy talking
    if (!_isSpeaking) {
      _isSpeaking = true;
      await _flutterTts.speak(text);
    }
  }

  void stop() {
    _flutterTts.stop();
  }
}

```

Figure 2: Configuring the TTS engine parameters

```

// Routing the detected pose to the active counter
void _routePoseToCounter(Pose pose) {
  switch (widget.exerciseName.toLowerCase()) {
    case 'pushups':
      _pushupCounter.processPose(
        pose,
        onRepCount: () => _ttsService.speak("Good repp!"),
        onFormError: () => _ttsService.speak("Error! Try again"),
      );
      break;
    case 'lateral raises':
      _lateralRaiseCounter.processPose(pose,
        onRepCount: () => _ttsService.speak("Good repp!"),
        onFormError: () => _ttsService.speak("Face the camera and raise your arms evenly"),
      );
      break;
    case 'bicep curls':
      _bicepCurlCounter.processPose(
        pose,
        onRepCount: () => _ttsService.speak("Good repp!"),
        // Elbow swinging error
        onFormError: () => _ttsService.speak("Try again! Keep your elbows in a fixed position"),
        // Only one arm used error
        onSymmetryError: () => _ttsService.speak("Use both arms while curling"),
      );
      break;
    default:
      _squatCounter.processPose(
        pose,
        onRepCount: () => _ttsService.speak("Good repp!"),
        onFormError: () => _ttsService.speak("Error! Try again"),
      );
  }
}

```

Figure 3: Conditional logic triggering auditory feedback based on real-time pose validation

Implementation:

To enhance the user's safety and focus, KasRrat utilizes a Text-to-Speech (TTS) system which is implemented using the flutter_tts plugin, which communicates via Asynchronous Method Channels to the device's native operating system (Android/iOS)

Workflow:

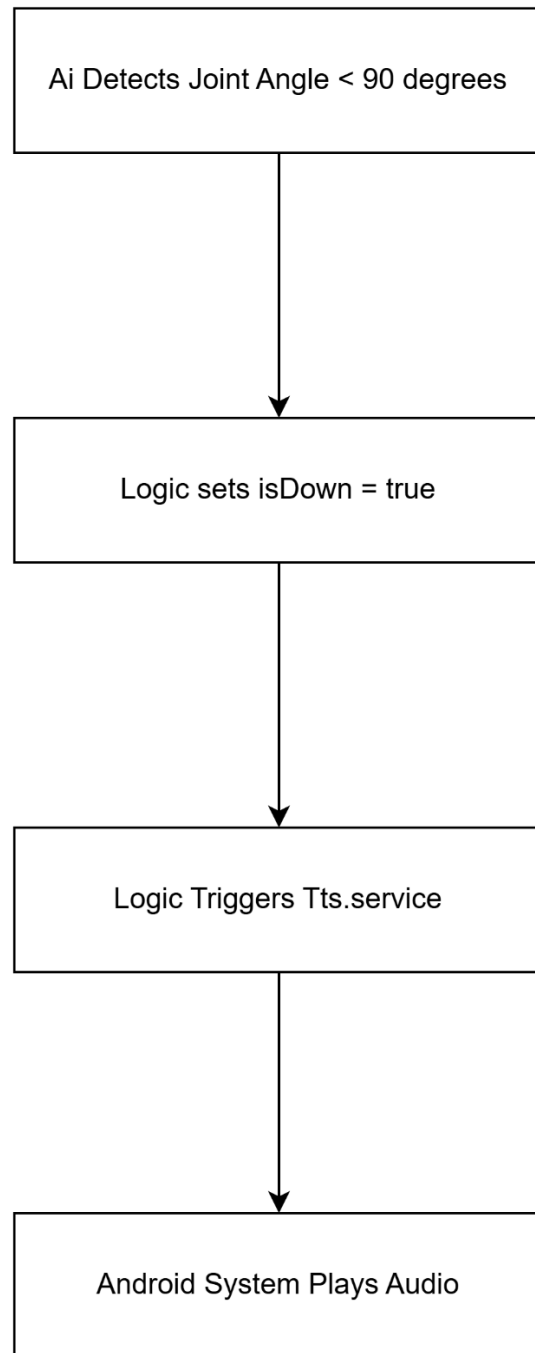


Figure 4: TTS Workflow